Assessing Sanctions Impact on Ransomware Payment Flows

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Executive Summary

- **Modern ransomware operations rely on cryptocurrency** for the extortion of victims.
    - Cryptocurrency provides a form of currency outside the direct control of any government or bank.
    - Mixer services and exchanges help operators convert money into fiat currency.
- We state with **medium confidence** that **sanctions disrupt the operations of malicious exchanges and mixers.**
- We state with **low confidence** that **sanctions disrupt the operations of ransomware operators** using these exchanges and mixers.
- The **duration and durability of these impacts is unknown**, due to lack of visibility into recovery times on actor operations.
- **Continuing to apply sanctions on identified criminal infrastructure is recommended.**

Methodology

We attempt to measure the change in payment flow from ransomware victims to sanctioned exchange and mixer services. For simplicity, we limited our analysis to the Bitcoin blockchain.

First, we build a corpus of publicly reported ransomware payment addresses and US-sanctioned mixer/exchange addresses. We ingest these addresses into a graph database analysis tool, called Synapse. We continually obtained additional transaction data for these addresses and newly discovered addresses via the Blockcypher API.

Next, we identify paths through the graph from payment addresses to sanctioned addresses. This is done through a modified application of the Breadth First Search algorithm. Once paths are identified, we attempt to perform analysis on the number and volume of transactions before and after sanctions for initial payment addresses, sanction addresses, and the paths from payment to sanction addresses.

Building a Corpus

We downloaded a JSON file detailing publicly reported ransomware payment addresses from the Ransomwhere project.[1] This contains data provided voluntarily by the victims of ransomware attacks. Data is vetted to ensure a reasonable degree of accuracy. We wrote Storm code (the language used by Synapse) to ingest the obtained JSON data into our graph database. After ingesting the data, it was decided to focus on only payments that had been verified as involving named ransomware families/actors, and we removed unaffiliated extortion addresses. A total of 10,465 addresses were obtained this way.[2]

We downloaded the SDN Entity List from the Office of Foreign Assets Control (OFAC) website.[3] We then used Python code to extract all Bitcoin addresses and their associated sanctions programs from the XML file into a CSV file.[4] Once this was done, all addresses sanctioned under the CYBER2 and CYBER3 programs were loaded into Synapse's graph and marked as sanctioned. The associated entity and time of sanction application were applied by hand afterwards. Any addresses associated with individuals were ignored. For additional context, we also manually added additional unsanctioned exchange addresses in the graph and marked them accordingly.[5]

Entity	**Address Count**	**Date of Sanction (YYYY/MM/DD)**
Hydramarket	117	2022/04/05
Blendrio	45	2022/05/06
Chatex	22	2021/11/08
Suex	14	2021/09/21
Zservers	3	2022/05/06

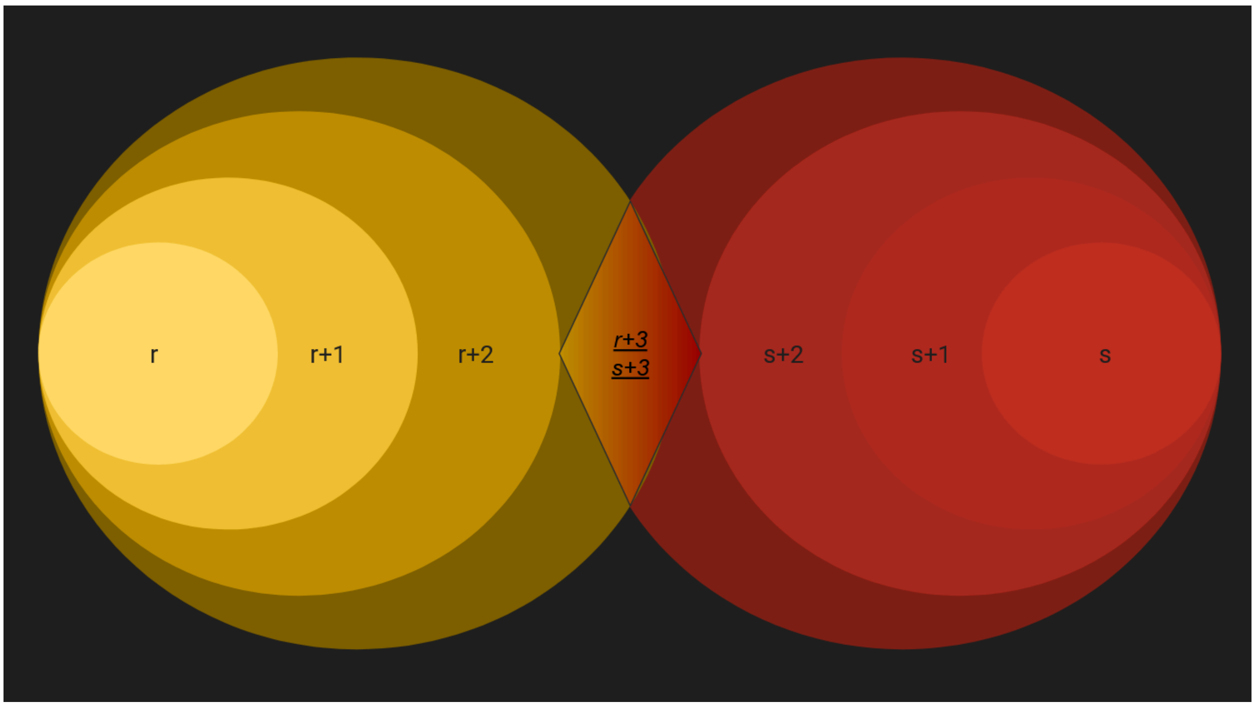
Identifying Payment Flows

Finding payment paths from an address to which a victim paid money to a sanctioned address is a graph traversal problem. We have a directed graph, with nodes representing

[1] Cable, Jack. (2024). Ransomwhere: A Crowdsourced Ransomware Payment Dataset (1.1.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.6512122
[2] See ransomware_data.json for the raw JSON data obtained from Ransomwhere. See ingest_ransomwhere_data.storm for the Storm code used to parse this data.
[3] SDN_ENHANCED.XML, located at https://sanctionslist.ofac.treas.gov/Home/SdnList
[4] See parse_sdn_list.py for the Python code used to parse the SDN list. See ofac_sanctions_w_programs.csv for the parsed data containing Bitcoin addresses and their associated sanctions program.
[5] See https://gist.github.com/f13end/bf88acb162bed0b3dcf5e35f1fdb3c17 for a list of the marked exchanges.

cryptocurrency addresses, and directed edges representing payments from one address to another.[6] The volume of data to sift through is too great for manual analysis.

One of the ways to search for a path through a graph is through the Breadth First Search (BFS) algorithm. Given a starting node, one visits all neighboring nodes. For each of those nodes, one visits their neighboring nodes, and so on until the desired node is found. While standard BFS is technically a viable way to find a path from a ransomware payment address to a sanctioned address, we rapidly run into a scaling problem in implementation.[7] To address this, we perform a process we have named "Converging Breadth First Search" (CBFS). The logic is demonstrated in the image below. The code used to implement this is present on GitHub.[8]



1. Mark all ransomware payment addresses with *r* and all sanctioned addresses with *s*
2. Traverse transactions to adjacent Bitcoin addresses as done in BFS, marking each address with $r+n$ and $s+n$ as we go, where n is the distance from the original ransomware or sanctioned address. In our implementation in Synapse, we marked each node with the address of the previous node as well (i.e. node $r+2$ above knows it is adjacent to $r+1$)

[6] Strictly speaking, Synapse's graph is actually a *hypergraph*, but we do not need complete accuracy for this analysis. See Synapse's documentation for a description of the difference between graph types.

[7] While we may not actually need to *visit* each of the exchange's nodes in a traditional BFS strategy, we will end up *storing* all of these nodes, which is of greater concern, as we are more limited in storage than in compute for this project.

[8] See mark_address_v3.storm for the Storm code used to implement Converging Breadth First Search. See identify_path.storm for the Storm code used to identify a path given a node marked with $r+n$ and $s+n$ as described above.

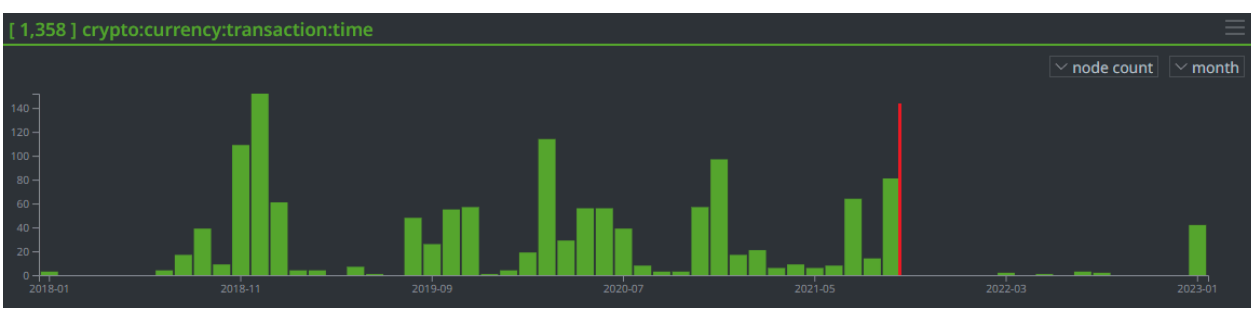
3. Once complete, search for nodes marked with both $r+n$ and $s+n$. Any node marked with both is a part of a path demonstrating the flow of Bitcoin from a ransomware to a sanctioned address

Comparison of worst-case number of nodes evaluated using BFS vs CBFS		
	BFS	**CBFS**
Scenario 1 *7 nodes each have 10 adjacent nodes*	$10^7 = 10,000,000$	$10^3 + 10^4 = 12,000$
Scenario 2 *3 nodes have 3 adjacent nodes* *2 nodes have 10 adjacent nodes* *1 node has 20 adjacent nodes* *1 node has 100 adjacent nodes*	$3^3 * 10^2 * 20^1 * 100^1 = 5,400,000$	$3^3 * 10^2 + 20^1 * 100^1 = 20,090$

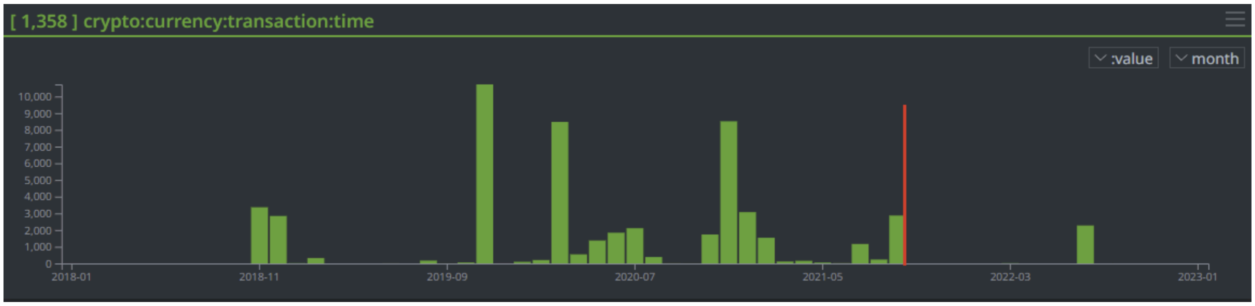
Analysis

Suex

Suex was sanctioned on 2021/09/21. We identified paths from known Conti ransomware payment addresses to Suex addresses, implying that Suex was the exchange of choice for use by the ransomware operators.



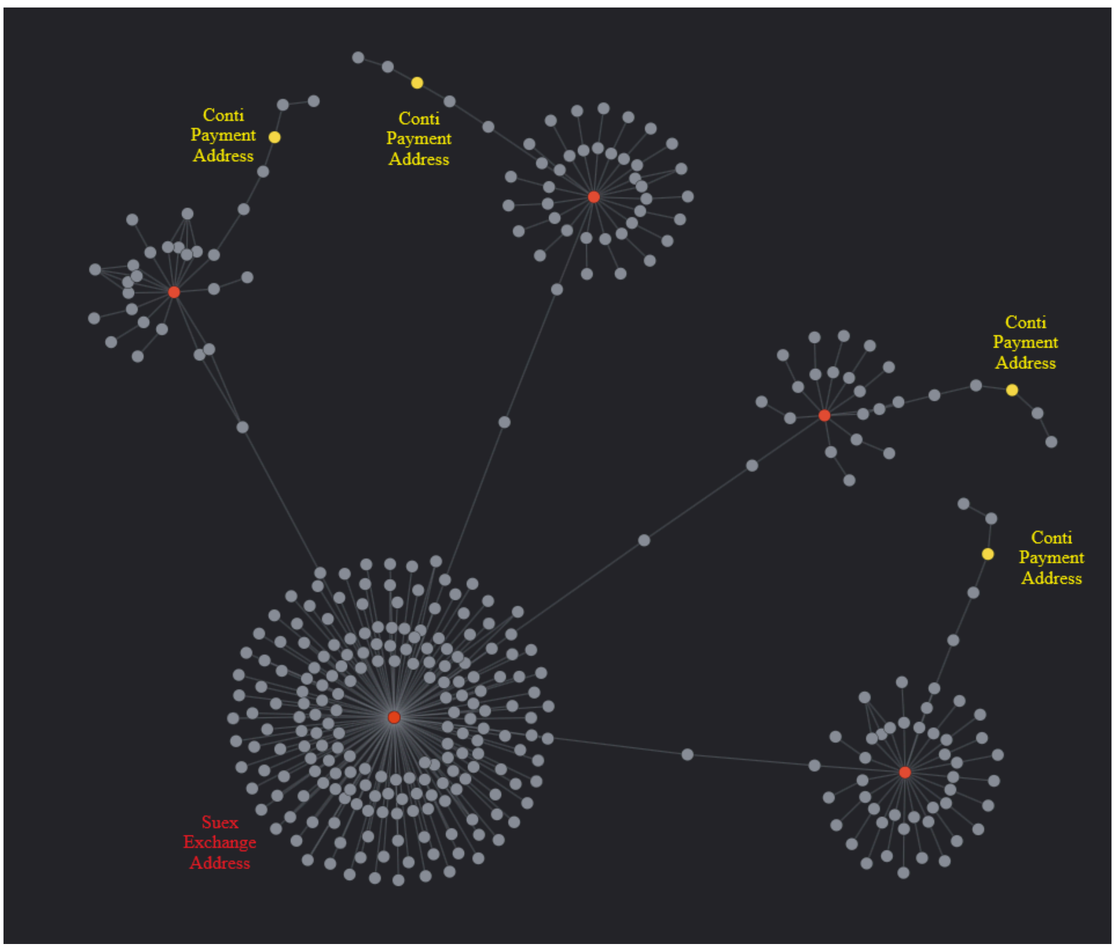
Number of Suex Transactions by Month (Sept 2018 - Nov 2021)



Volume (BTC) of Suex Transactions by Month

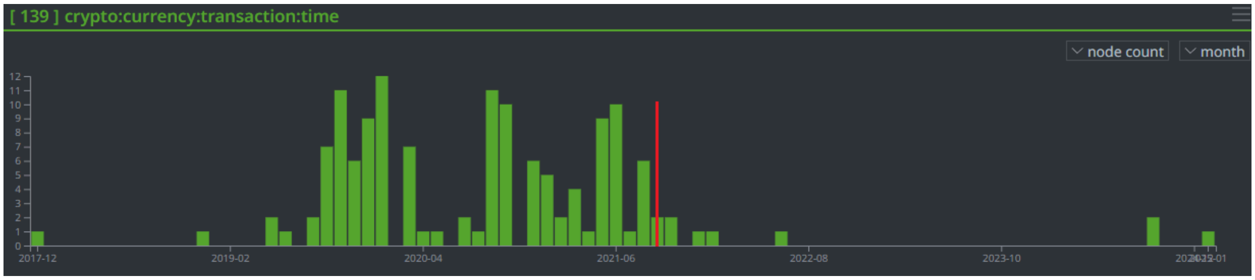
The decrease in activity following the sanctions is notable in the graph above, demonstrating the impact of sanctions. However, we can show that the impact may have successfully extended to ransomware operators.

In examining the paths identified, we found that addresses associated with the Conti ransomware family sent bitcoin through the blockchain to addresses associated with Suex, depicted below.

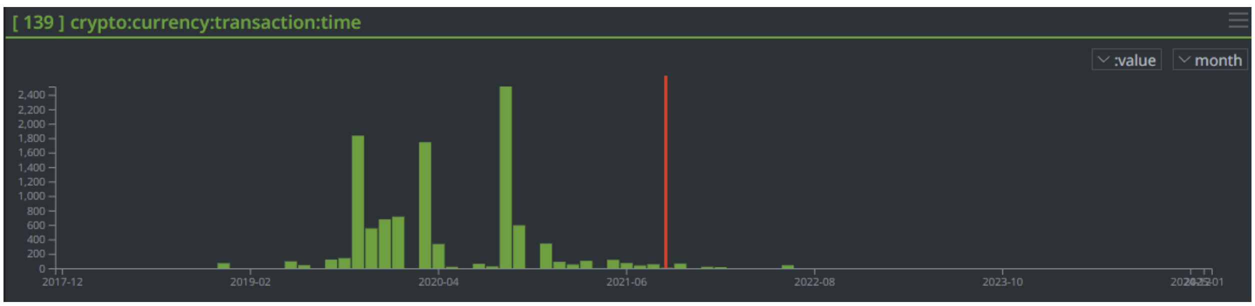


Force Graph Depiction of Paths from Conti Payment Addresses to Suex Exchange Addresses

Examining the frequency and volume of payments to known Conti ransomware addresses, we found that there was a sizable decrease in payment frequency and volume following the sanctions applied to Suex. This suggests that the sanctions placed on Suex not only impacted the exchange, but also impacted the ransomware operators relying on it.



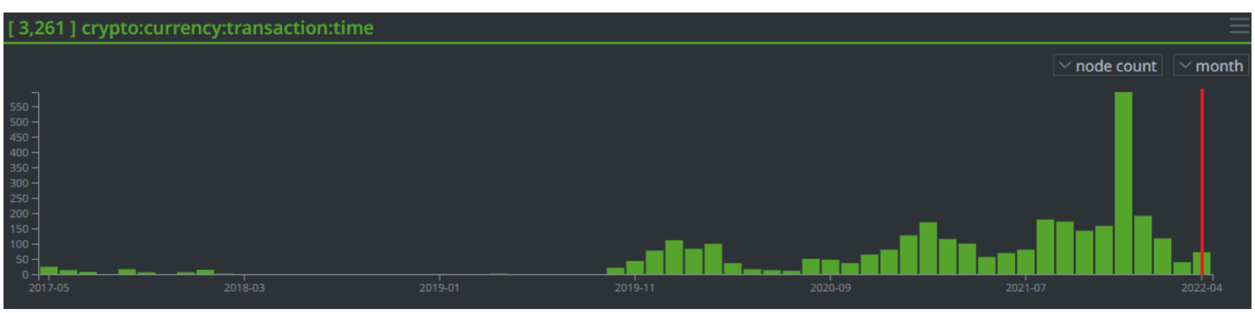
Number of Conti Transactions by Month (Dec 2017 - Jan 2025)



Volume (BTC) of Conti Transactions by Month

Hydramarket

Hydramarket, one of the largest darknet markets for illegal transactions at the time, was sanctioned on 2022/04/05.[9] Examining the payments received or sent by the sanctioned addresses shows a significant decrease in activity following the application of sanctions.



Number of Hydramarket Transactions by Month (May 2018 - April 2022)

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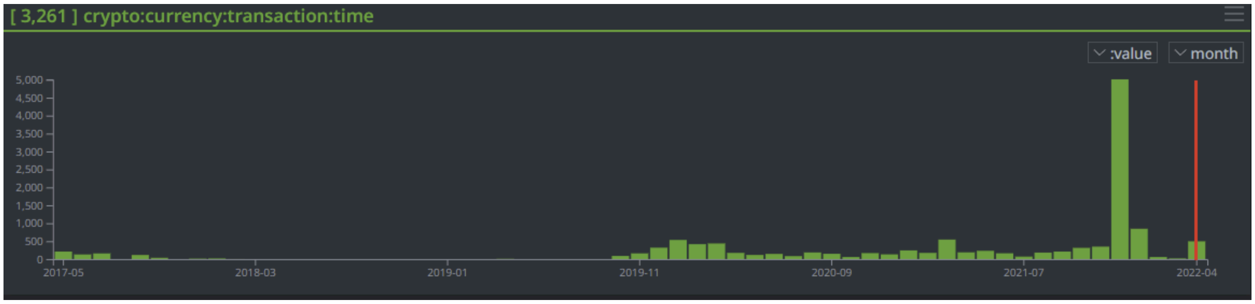
[9] See the Department of Justice Press Release and the Department of the Treasury Press Release.



Volume (BTC) of Hydramarket Transactions by Month

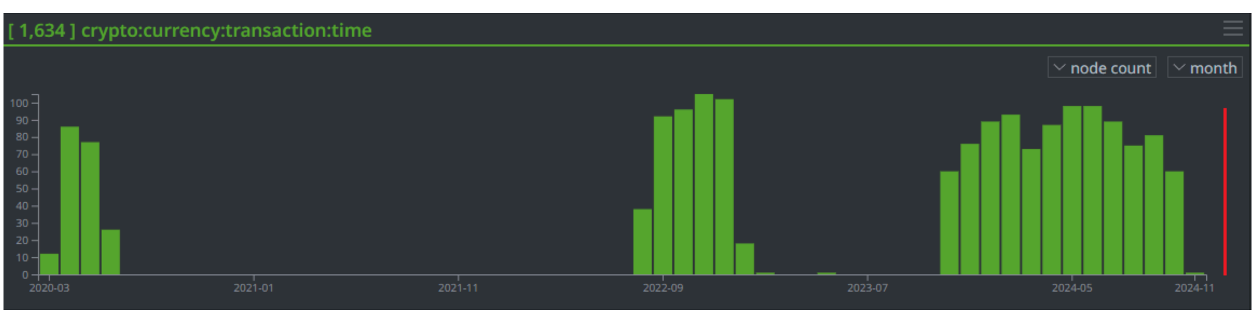
At first glance, these sanctions appear to have clearly impacted Hydramarket. However, there are important caveats to consider. First, the servers were seized by German authorities. This may have destroyed the *technical capability* to transact with Hydramarket, rendering the sanctions irrelevant. Second, no Hydramarket administrators were arrested, so it remains possible that the site could simply be recreated under another name by the same operators.[10]

We were unable to confidently identify multiple paths to Hydramarket from sets of ransomware families. The reason for this is that many such paths appear to go through **~~redacted~~** addresses.[11] The sheer volume of transactions made it difficult to confirm any paths that our existing Storm code produced, and additional work is needed to return the relevant transactions for verification (as opposed to simply returning the addresses involved).[12]

Zservers

Zservers, a hosting provider used by the LockBit ransomware group, was sanctioned on 2025/02/11.[13] The impact of sanctions on Zservers is less clear than other sanctioned entities.

There are three associated Bitcoin addresses sanctioned by OFAC. Each address corresponds almost exactly to one of the three visible periods of activity shown on the charts below.



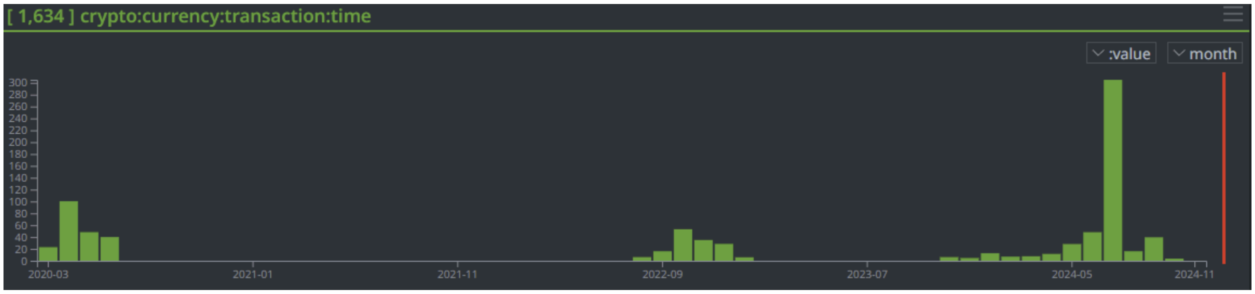
[10] See https://www.bbc.com/news/technology-61002904 for information on the Hydramarket takedown

[11] **~~redacted~~**.

[12] **~~redacted~~**.

[13] See the Department of the Treasury Press Release.

Number of Zservers Transactions by Month (Mar 2020 - Dec 2024)

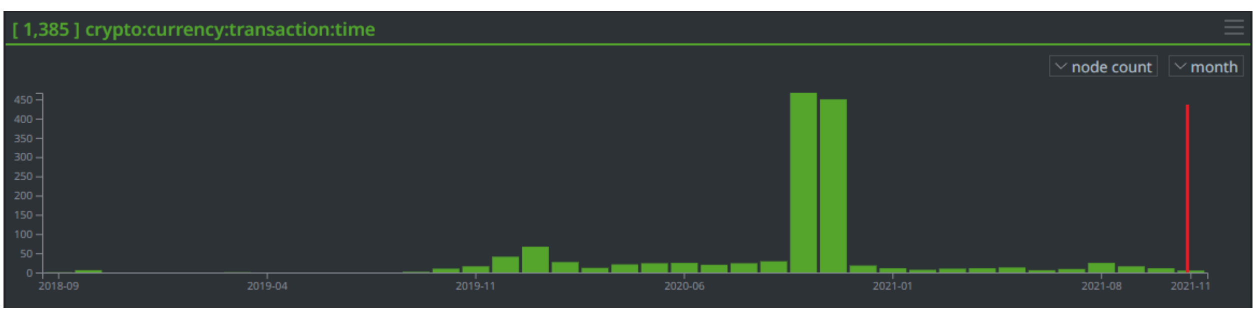


Volume (BTC) of Zservers Transactions by Month

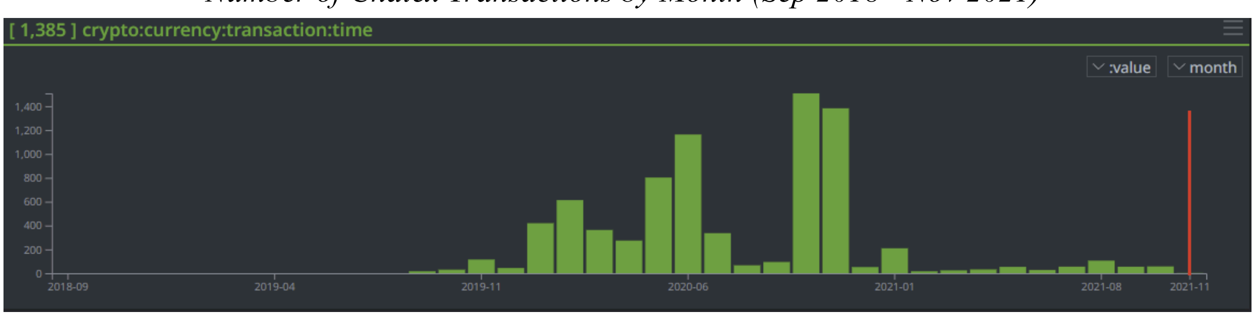
Transactions to Zservers appear to have slowed dramatically in November 2024, and no transactions were present for any of these addresses in December before the application of sanction. Combined with the gaps in activity displayed in the charts above, this makes it difficult to assess whether or not sanctions are impacting Zservers. Once again, verification of the ransomware families making use of Zservers was unsuccessful, due to the presence of **~~redacted~~** in the identified paths.

Chatex

Chatex was sanctioned on 2021/11/08. We observed a significant drop in activity roughly a year before sanctions were applied. However, activity ceased completely within one month of the application of sanctions. We consider it plausible that sanctions may have contributed to the end of this activity, but this case does not provide strong evidence by itself.



Number of Chatex Transactions by Month (Sep 2018 - Nov 2021)



Volume (BTC) of Chatex Transactions by Month

Conclusion

Findings & Limitations

- ***Finding:*** **Sanctions appear to effectively reduce traffic to exchanges and mixers** used by ransomware operators in most cases.
- ***Finding:*** These effects also appear to flow upstream, showing that the **sanctions impact the ransomware operators themselves**.
- ***Limitation:*** **The duration of this impact is unclear**. Attribution of cybercrime can be challenging, and it remains unclear whether impacted infrastructure is simply rebuilt under new names or if ransomware operators change their "branding" and used tools, both of which would not show up in this analysis.
- ***Limitation:*** This report was hindered by compute/storage limitations for Synapse and API rate limiting from Blockcypher. Both of these mean that **this report only analyzed a subset of all transactions that may have occurred** between the sanctioned infrastructure and the ransomware payment addresses.

Recommendations

- **Continue to sanction the services used by ransomware operators** - these sanctions do appear to have an impact
- **Continue searching for a long-term solution to ransomware**. We have demonstrated that sanctions may suppress ransomware attacks, but they do not raise the cost of criminal activity enough to discourage the ransomware industry en masse. **Sanctions are a short-term solution**.

Citations

Cable, Jack. (2024). Ransomwhere: A Crowdsourced Ransomware Payment Dataset (1.1.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.6512122

f13end, "Wallet Addresses." GitHub Gist. https://gist.github.com/f13end/bf88acb162bed0b3dcf5e35f1fdb3c17

"Graphs and Hypergraphs," Synapse User Guide. The Vertex Project, LLC. https://synapse.docs.vertex.link/en/stable/synapse/userguides/background.html#graphs-and-hypergraphs

Tidy, Joe. "Hydra: How German police dismantled Russian darknet site." April 5, 2022. *BBC*, https://www.bbc.com/news/technology-61002904

U.S. Department of Justice. April 5, 2022. *Justice Department Investigation Leads to Shutdown of Largest Online Darknet Marketplace*. https://www.justice.gov/archives/opa/pr/justice-department-investigation-leads-shutdown-largest-online-darknet-marketplace

U.S. Department of the Treasury. February 11, 2025. *United States, Australia, and the United Kingdom Jointly Sanction Key Infrastructure that Enables Ransomware Attacks*. https://home.treasury.gov/news/press-releases/sb0018

U.S. Department of the Treasury. April 5, 2022. *Treasury Sanctions Russia-Based Hydra, World's Largest Darknet Market, and Ransomware-Enabling Virtual Currency Exchange Garantex.* https://home.treasury.gov/news/press-releases/jy0701

U.S. Department of the Treasury, Office of Foreign Assets Control. *OFAC Sanctions List Service.* https://sanctionslist.ofac.treas.gov/Home/SdnList

Appendix A: Verified Conti Payment Paths[14]

Chain 1

Type	Identifier	Description
Address	3HWbmfqVFxdLJgDHTtDPoyswGbxZJZwYNk	Payment Address
Transaction	ee69c277ab29d2cce04c7154778655b5cd7648450c3ef741943c4b29e18bcb54	
Address	1PYwNUuiJ3SQFqmNsA9Ys3HVaYtGG6fe8G	Transit Address
Transaction	594999e9736e55b5e1590cbf11d2f649b5dff5718836e7f7c5da81ce11345c61	
Address	12HQDsicffSBaYdJ6BhnE22sfjTESmmzKx	Sanctioned Address / Suex Exchange

Chain 2

Type	Identifier	Description
Address	1KUzq13qqefyBoHtiS7J8E9uHhZiRwgxvM	Payment Address
Transaction	d7d9c270071b6bf12e6befbe6b117560847a8db4d506ee07d58aaddc3aab397d	
Address	1M1A5WM4d2wrLCZJCsgHse7ZSMz7hkk1aN	Transit Address
Transaction	47a3d8ebd6e90787ff8bb6143c9260b01f7f70903c23b3a6fa750cda2b33fc50	
Address	12HQDsicffSBaYdJ6BhnE22sfjTESmmzKx	Sanctioned Address / Suex Exchange

Chain 4

Type	Identifier	Description
Address	36KQWiUJtZsTiezhbPHUBUEo4JBhxCWmkM	Payment Address

[14] "Chain 3" is omitted, due to a technical error. The nodes in the graph making up this path actually represent the sanctioned address and the ransomware payment address sending Bitcoin to a common address, rather than the currency flowing from the payment address to the sanctioned address.

Transaction	ec9a8b9a72a3d1f9bef2f96b7c0fa3f65d0a5a1aca5ca0209dd700db1ca7edd0	
Address	1H2rU6BeTrzw6WqxGsNHKS2GT6LP7JKVvM	Transit Address
Transaction	7e3853540d2d1acb8c314d80447aa6aebeff323f591fd85685c1d6cec1eca664	
Address	12HQDsicffSBaYdJ6BhnE22sfjTESmmzKx	Sanctioned Address / Suex Exchange

Chain 5

Type	Identifier	Description
Address	bc1q69htg0wx5vafqmgusxrvgf6cakfd0na6r89ydh	Payment Address
Transaction	eb986b6b38ecb9c7fed92d0b4f727a212a04851e21f995e8a2077290e71c7136	
Address	1HPrZXKrYsjfH2TSHzkHqrLcYoZ8AGzS4v	Transit Address
Transaction	53732ec4895aa90b03a973a10d6ac53c5493c0fd4a575c6577255ac80fef108b	
Address	12HQDsicffSBaYdJ6BhnE22sfjTESmmzKx	Sanctioned Address / Suex Exchange

Appendix B: Code

Most code used can be viewed in this GitHub repository, SEST-6650-Ransomware-Sanctions-Effectiveness.

Code	Description
enrich_address_blockcypher.storm	Used to obtain additional address and transaction data from Blockcypher's API and model it directly in Synapse's graph.
identify_path.storm	Used to automatically identify a path from a ransomware payment address to a sanction address. Must be executed on a node marked with tags from *mark_address_v3.storm*. Naive implementation, requires manual verification.
ingest_ofac_sanctions_list.storm	Used to load Bitcoin addresses from a file into Synapse's graph.
ingest_payment_addresses_w_no_transactions.storm	Used to take the addresses identified in *blockchain_com.py* and mark them as possessing no transaction data. Needed for data filtering and to address limitations in the blockcypher enrichment process.
ingest_ransomwhere_data.storm	Used to load addresses from the Ransomwhere dataset into Synapse's graph and mark them according to their associated ransomware families.
macro_inbound_transaction_addresses.storm	Used to simplify pivoting in Synapse's graph. Given a cryptocurrency address node, identifies any other cryptocurrency address nodes which have sent Bitcoin to this one.
macro_outbound_transaction_addresses.storm	Used to simplify pivoting in Synapse's graph. Given a cryptocurrency address node, identifies any other cryptocurrency address nodes to which this one has sent Bitcoin.
mark_address_v3.storm	The Storm code that implements the Converging Breadth First Search marking technique described in the report.
blockchain_com.py	Used to identify ransomware payment addresses from the Ransomwhere dataset which have *not* been observed involved in any transactions. Used for data filtering and to address limitations in the blockcypher enrichment process.
parse_sdn_list.py	Used to identify Bitcoin addresses sanctioned by OFAC and their associated sanctions program.